

Jeffrey D. '84 and Karen G. '82 Camm Lecture

The Jeffrey D. '84 and Karen G. '82 Camm Lecture brings distinguished scholars to Clemson to highlight impactful applications of the mathematical and statistical sciences and to celebrate faculty excellence.



New Opportunities in Fake Data

Barry L. Nelson

Walter P. Murphy Professor Emeritus
Department of Industrial Engineering and Management Sciences
Northwestern University

October 14, 2026

11:15 AM–12:05 PM

Watt Family Innovation Center Auditorium

About the Talk

Stochastic computer simulation—literally, the generation of “fake data”—is a practical tool for the design and improvement of dynamic systems that are subject to uncertainty. Research in and application of stochastic simulation sits at the interface of industrial engineering/operations research, computer science, and statistics.

Ubiquitous parallel computing and the need for system control through digital twins have created new opportunities that require new ways of thinking about simulation. This talk introduces two of these opportunities through real applications. The presentation is math-light and begins with an audience-participation simulation that demonstrates the key ideas.

About the Speaker

Barry L. Nelson is the Walter P. Murphy Professor Emeritus in the Department of Industrial Engineering and Management Sciences at Northwestern University. His research focuses on the design and analysis of computer simulation experiments for discrete-event stochastic systems, including simulation optimization, model risk, variance reduction, output analysis, metamodeling, and multivariate input modeling.

His applications span manufacturing, services, financial engineering, renewable energy generation, and transportation. He has published numerous papers and three books, including *Foundations and Methods of Stochastic Simulation: A First Course* (Springer, 2nd ed., 2021). Nelson is a Fellow of INFORMS and IISE and has received major honors from both organizations for research, simulation, expository writing, and teaching.

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[Barry Nelson's website](#)